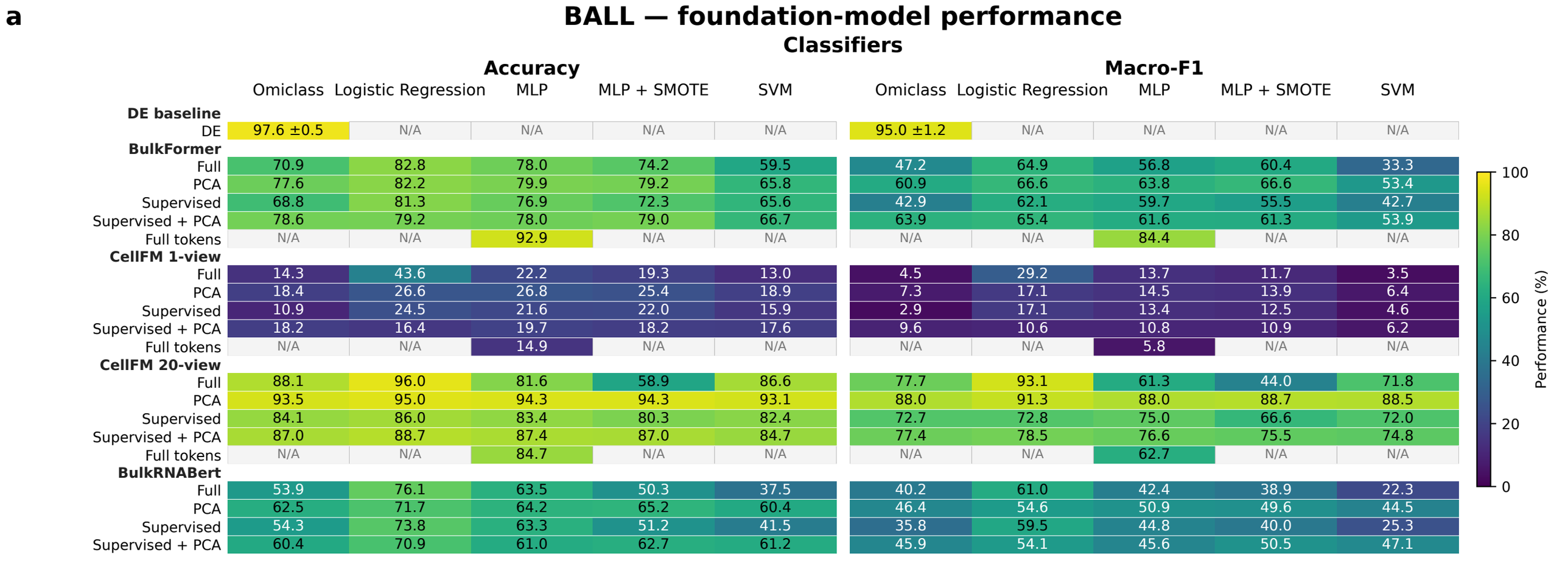
a
BALL — foundation-model performance
Classifiers
Accuracy
Macro-F1
Omiclass
Logistic Regression
MLP
MLP + SMOTE
SVM
DE baseline
DE
97.6 ±0.5
95.0 ±1.2
BulkFormer
Full
70.9
82.8
78.0
74.2
59.5
47.2
64.9
56.8
60.4
33.3
PCA
77.6
82.2
79.9
79.2
65.8
60.9
66.6
63.8
66.6
53.4
Supervised
68.8
81.3
76.9
72.3
65.6
42.9
62.1
59.7
55.5
42.7
Supervised + PCA
78.6
79.2
78.0
79.0
66.7
63.9
65.4
61.6
61.3
53.9
Full tokens
92.9
84.4
CellFM 1-view
Full
14.3
43.6
22.2
19.3
13.0
4.5
29.2
13.7
11.7
3.5
PCA
18.4
26.6
26.8
25.4
18.9
7.3
17.1
14.5
13.9
6.4
Supervised
10.9
24.5
21.6
22.0
15.9
2.9
17.1
13.4
12.5
4.6
Supervised + PCA
18.2
16.4
19.7
18.2
17.6
9.6
10.6
10.8
10.9
6.2
Full tokens
14.9
5.8
CellFM 20-view
Full
88.1
96.0
81.6
58.9
86.6
77.7
93.1
61.3
44.0
71.8
PCA
93.5
95.0
94.3
94.3
93.1
88.0
91.3
88.0
88.7
88.5
Supervised
84.1
86.0
83.4
80.3
82.4
72.7
72.8
75.0
66.6
72.0
Supervised + PCA
87.0
88.7
87.4
87.0
84.7
77.4
78.5
76.6
75.5
74.8
Full tokens
84.7
62.7
BulkRNABert
Full
53.9
76.1
63.5
50.3
37.5
40.2
61.0
42.4
38.9
22.3
PCA
62.5
71.7
64.2
65.2
60.4
46.4
54.6
50.9
49.6
44.5
Supervised
54.3
73.8
63.3
51.2
41.5
35.8
59.5
44.8
40.0
25.3
Supervised + PCA
60.4
70.9
61.0
62.7
61.2
45.9
54.1
45.6
50.5
47.1
N/A
Performance (%)
100
80
60
40
20
0